

Aperiodically intermittent control for synchronization of switched complex networks with unstable modes via matrix ω -measure approach

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Abstract The study, by using aperiodically intermittent pinning control, is to synchronize switched delayed complex networks with unstable subsystems. Matrix ω -measure and mode-dependent average dwell time method are used to achieve globally exponential synchronization for such system. By designing the useful switching rule and control scheme, we obtain the novel

synchronization criteria, which improve the conventional results. Finally, simulation analysis demonstrates the advantages of proposed innovations.

Keywords Globally exponential synchronization (GES) · Switched complex networks · Aperiodically intermittent pinning control (AIPC) · ω -Measure · Mode-dependent average dwell time (MDADT)

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1 Introduction

Complex networks play a significant role in modeling many practical systems [1–4]. As a representative phenomenon, synchronization problem has drawn considerable attention recently owing to the comprehensive applications on image processing, secure communication, information science and automatic control [5–9]. Nowadays, some synchronization modes have been introduced and intensively investigated; for instance, many results with regard to cluster synchronization, complete synchronization, finite-time synchronization and projective synchronization have been derived in the past decades [10–13].

In many cases, networks synchronization cannot be reached by itself. Therefore, impulsive control [14–16], pinning control [17–19], sliding mode technique [20–22] as well as some advanced techniques [23–25] were employed to actuate the dynamical networks to objective state. As a discontinuous control scheme, intermittent control [26–30] has many advantages of reducing

the amount of transmitted information and economic applicability, since the intermittent feedback controller is active in the control time and interdictory in the recumbent time. However, the periodicity condition of intermittent control in most of the previous results may be unreasonable and restricted to some extent. Moreover, the pinning control strategy with feedback controller imposed only on a small proportion of the interconnected nodes is more applicable in reality, especially for large-scale networks. As a result, aperiodically intermittent pinning control (AIPC) can be utilized to synchronize complex networks.

In [29], Liu et al. constructed a novel differential inequality to synchronize delayed complex networks based on aperiodically intermittent pinning control. Wang considered hybrid coupling terms to synchronize complex network with mixed time delays by designing the applicable aperiodically intermittent pinning controller [30]. It is obvious that the classic Lyapunov function method has been employed. Recently, matrix measure approach adequately using elements of a matrix has become an efficient method to stabilize or synchronize complex networks without constructing a complicated Lyapunov function [31, 32]. Matrix measure approach relaxes the limitations of nonnegative constants of algebra and norm methods and, consequently, has broader application in practice. Up to now, no investigations on networks synchronization use AIPC and matrix measure approach.

As a typical hybrid dynamic system, switched complex network, which is composed of a limited amount of subsystems and a logic switching rule, has been concerned and applied successfully in varieties of fields including high-speed signal processing, artificial intelligence, flight control system and network control system [33–36]. So far, numerous significant results with respect to switched systems have been derived; in particular, the stability or synchronization problems of switched networks and the design of switching signal have become the focus of research [37, 38]. As the emblematical research technique, average dwell time (ADT) approach is efficient to stabilize and synchronize switched systems [39–41]. Moreover, MDADT approach which allows each subsystem to have its own specific average dwell time is verified to be less restricted and more feasible compared with ADT technique [42, 43].

It is worth noticing that switched systems containing unstable modes are normally inevitable due to con-

troller failures or sensor faults [44–47]. Furthermore, it is unnecessary to make all the subsystems be stable for practical large-scale complex networks. Therefore, it is meaningful to consider switched dynamical networks with unstable modes by utilizing MDADT. A brief review on the previous results indicates that Multiple Lyapunov Functions (MLFs) were considered to stabilize or synchronize the switched systems with unstable modes [44, 47]. However, it may be difficult to establish all positive definite Lyapunov functions for each subsystem of switched networks. Recently, matrix measure approach has become an effective method to synchronize switched complex networks without constructing complicated MLFs [48, 49]. In [49], Li et al. introduced matrix measure method to synchronize switched interval networks with stable subsystems undergoing arbitrary switching law. However, no related results concern the synchronization of switched complex networks with unstable modes via matrix measure approach and MDADT.

In practice, due to the long distance between different nodes and signal transmission crowding, time delay is inevitable with regard to many practical complex networks, and it may lead to oscillation behavior or network instability. Generally speaking, internal time delay that existed inside the dynamical network and coupling time delay that resulted from information transmission among different nodes are two typical cases [50]. In consequence, it is necessary to take different types of time delays into account to synchronize switched dynamical networks.

Motivated by the above-mentioned discussions, GES of switched delayed complex networks (SDCNs) with unstable modes will be investigated. The main innovations are highlighted as follows. Above all, matrix measure approach is utilized for the first time to synchronize SDCNs under AIPC, and mixed time-varying delays are considered. Additionally, we introduce the AIPC strategy to SDCNs with unstable modes and generalize some previous results. Thirdly, without constructing complicated MLFs, new criteria are derived to synchronize SDCNs based on ω -measure approach and MDADT.

Notations: A^T denotes the transpose of A . R^n is n -dimensional Euclidean space. $R^{m \times n}$ is $m \times n$ -dimensional real matrices. I_n is an n -dimensional identity matrix.

2 System model and preliminaries

The following complex networks composed of N identical nodes are considered:

$$\begin{aligned}\dot{x}_i(t) = & \tilde{f}(x_i(t)) + c_1 \sum_{j=1}^N d_{ij} \Gamma_1 x_j(t) \\ & + c_2 \sum_{j=1}^N a_{ij} \Gamma_2 x_j(t - \tau_2(t)) \\ & + c_3 \sum_{j=1}^N b_{ij} \Gamma_3 \int_{t-\tau_3(t)}^t x_j(v) dv + \tilde{\theta}_i(t), \quad (1)\end{aligned}$$

where $x_i(t) = (x_i^1(t), x_i^2(t), \dots, x_i^n(t))^T \in R^n$; $\tilde{f}(\cdot)$ is a continuously nonlinear function depicting the dynamic behavior of nodes and $\tilde{f}(x_i(t)) = f(t, x_i(t), x_i(t - \tau_1(t)))$ for any $f(\cdot)$ and $x_i(t)$. $0 \leq \tau_i(t) \leq \tau$ ($i = 1, 2, 3$), respectively, represent the internal, discrete and distributed delays. $c_i > 0$ ($i = 1, 2, 3$) are corresponding coupling strength. $0 < \Gamma_i \in R^{n \times n}$ ($i = 1, 2, 3$) represent the positive definite inner connecting matrices and satisfy $\Gamma_i = \Gamma_i^T$. $D = (d_{ij}) \in R^{N \times N}$, $A = (a_{ij}) \in R^{N \times N}$ and $B = (b_{ij}) \in R^{N \times N}$ are coupling matrices. Moreover, D , A and B satisfy $d_{i1} + d_{i2} + \dots + d_{iN} = 0$, $a_{i1} + a_{i2} + \dots + a_{iN} = 0$, $b_{i1} + b_{i2} + \dots + b_{iN} = 0$. By setting all mentioned delays, we consider the initial conditions as $x_i(v) = \psi_i(v) \in C([- \tau, 0], R^n)$. And $\tilde{\theta}_i(t)$ is the aperiodically intermittent pinning feedback controller to be designed.

Remark 1 Time-varying delays in (1) do not have other restrictions but the bounded condition. Moreover, the coupling configuration matrices D , A and B do not claim to be symmetric or irreducible. Hence, the proposed model (1) is more general and consistent with reality.

For the conciseness of formulation, (1) is expressed as

$$\begin{aligned}\dot{x}(t) = & \tilde{f}(x(t)) + c_1(D \otimes \Gamma_1)x(t) \\ & + c_2(A \otimes \Gamma_2)x(t - \tau_2(t)) \\ & + c_3(B \otimes \Gamma_3) \int_{t-\tau_3(t)}^t x(v) dv + \tilde{\theta}(t), \quad (2)\end{aligned}$$

where $x(t) = (x_1(t), \dots, x_N(t))^T$, $\tilde{\theta}(t) = (\tilde{\theta}_1(t), \dots, \tilde{\theta}_N(t))^T$. Suppose the equilibrium point $\varphi(t)$ to be

$$\dot{\varphi}(t) = f(t, \varphi(t), \varphi(t - \tau_1(t))). \quad (3)$$

Then AIPC scheme is proposed as

$$\tilde{\theta}(t) = \begin{cases} -U(x(t) - \varphi(t)), & t \in [t_m, h_m), \\ 0, & t \in [h_m, t_{m+1}), \end{cases} \quad (4)$$

where $[t_m, t_{m+1})$ represents an aperiodically intermittent control period, of which $[t_m, h_m)$ is the control time and $[h_m, t_{m+1})$ is the rest time, $m = 0, 1, 2, \dots$. The control gain matrix with l pinning nodes is denoted by $U = \text{diag}(u_1, u_2, \dots, u_l, 0, \dots, 0) \in R^{N \times N}$. The error is to be $\tilde{h}(t) = x(t) - \varphi(t)$, and then we have

$$\begin{cases} \dot{\tilde{h}}(t) = \tilde{H}(\tilde{h}(t)) + c_1(D \otimes \Gamma_1)\tilde{h}(t) \\ \quad + c_2(A \otimes \Gamma_2)\tilde{h}(t - \tau_2(t)) \\ \quad + c_3(B \otimes \Gamma_3) \int_{t-\tau_3(t)}^t \tilde{h}(v) dv \\ \quad - U\tilde{h}(t), & t \in [t_m, h_m), \\ \dot{\tilde{h}}(t) = \tilde{H}(\tilde{h}(t)) + c_1(D \otimes \Gamma_1)\tilde{h}(t) \\ \quad + c_2(A \otimes \Gamma_2)\tilde{h}(t - \tau_2(t)) \\ \quad + c_3(B \otimes \Gamma_3) \int_{t-\tau_3(t)}^t \tilde{h}(v) dv, & t \in [h_m, t_{m+1}), \end{cases} \quad (5)$$

where $\tilde{H}(\tilde{h}(t)) = \tilde{f}(x(t)) - \tilde{f}(\varphi(t))$. In light of (1), the switched delayed complex networks with unstable modes are represented as

$$\begin{aligned}\dot{x}_i(t) = & \tilde{f}_{\sigma(t)}(x_i(t)) + c_{1\sigma(t)} \sum_{j=1}^N d_{ij\sigma(t)} \Gamma_{1\sigma(t)} x_j(t) \\ & + c_{2\sigma(t)} \sum_{j=1}^N a_{ij\sigma(t)} \Gamma_{2\sigma(t)} x_j(t - \tau_2(t)) \\ & + c_{3\sigma(t)} \sum_{j=1}^N b_{ij\sigma(t)} \Gamma_{3\sigma(t)} \int_{t-\tau_3(t)}^t x_j(v) dv \\ & - U_{\sigma(t)}(x_i(t) - \varphi(t)), \quad (6)\end{aligned}$$

where $\sigma(t) : [0, +\infty) \rightarrow \Lambda = \{1, 2, \dots, Q\}$. For $t_0 < t_1 < t_2 < \dots < t_k < t_{k+1} < \dots$, we suppose $\sigma(t_k)$ th subsystem to be activated within $t \in [t_k, t_{k+1})$, and the instantaneous value of the switching rule σ is available in practice. Moreover, we consider that Q subsystems can be stable or unstable. And we also assume that there exists ι ($1 \leq \iota \leq Q$) stable subsystems and $Q - \iota$ unstable subsystems and denote $R = \{1, 2, \dots, \iota\}$ and $T = \{\iota + 1, \iota + 2, \dots, Q\}$; R and T are the stable subsystems sets and the unstable subsystems sets, respectively.

Some preliminaries can be shown as follows. First, the assumptions are given as

Assumption 1 When the scalars z_{1i} and z_{2i} are existing, then for $t \geq 0$, $\forall y_i(t), \varphi_i(t) \in R^n$,

$$\begin{aligned}\tilde{f}(y_i(t)) - \tilde{f}(\varphi_i(t)) \leq & z_{1i}(y_i(t) - \varphi_i(t)) \\ & + z_{2i}(y_i(t - \tau_1(t)) - \varphi_i(t - \tau_1(t))).\end{aligned}$$

Assumption 2 There exist $0 < \delta < \varrho < +\infty$, such that for any $m = 0, 1, 2, \dots$, the aperiodically intermittent control strategy satisfies:

$$\begin{cases} \inf_m (h_m - t_m) = \delta > 0, \\ \sup_m (t_{m+1} - t_m) = \varrho < +\infty, \end{cases} \quad (7)$$

and we define

$$\theta = \lim_{m \rightarrow +\infty} \sup \frac{t_{m+1} - h_m}{t_{m+1} - t_m}, \quad (8)$$

where $\theta \in (0, 1)$.

Definition 1 [31] The ω -measure of $W = (w_{ij}) \in R^{N \times N}$ is $\mu_\omega(W) = \lim_{\chi \rightarrow 0^+} \frac{\|I_N + \chi W\|_\omega - 1}{\chi}$, where I_N is an identity matrix; $\|W\|_\omega = \max_j \sum_{i=1}^N \frac{\omega_i}{\omega_j} |w_{ij}|$ and $\omega_i > 0$ are arbitrary constants. By some calculations, we have $\mu_\omega(W) = \max_j \{w_{jj} + \sum_{i=1, i \neq j}^N \frac{\omega_i}{\omega_j} |w_{ij}|\}$.

Definition 2 [43] For $\sigma(t)$, $\forall t_2 > t_1 > 0$ and $\forall p \in \Lambda$, let $N_{\sigma p}(t_2, t_1)$ and $T_p(t_2, t_1)$ denote the numbers of switching and the total operation time of the p th subsystem on $[t_1, t_2]$, respectively. Then, τ_{ap} are called MDADT if there exists positive scalar N_{0p} such that

$$N_{\sigma p}(t_2, t_1) \leq N_{0p} + \frac{T_p(t_2, t_1)}{\tau_{ap}}.$$

Lemma 1 [30] For $t \in [-\tau, \infty)$, suppose continuous function $\tilde{h}(t)$ satisfies

$$\begin{cases} \dot{\tilde{h}}(t) \leq -b_1 \tilde{h}(t) + b_3 \sup_{t-\tau \leq v \leq t} \tilde{h}(v) \\ \quad + b_4 \int_{t-\phi(t)}^t \tilde{h}(v) dv, \quad t_m \leq t < h_m, \\ \dot{\tilde{h}}(t) \leq b_2 \tilde{h}(t) + b_3 \sup_{t-\tau \leq v \leq t} \tilde{h}(v) \\ \quad + b_4 \int_{t-\phi(t)}^t \tilde{h}(v) dv, \quad h_m \leq t < t_{m+1}, \end{cases}$$

where b_1, b_2, b_3, b_4 are positive constants, $\phi(t) \leq \tau$, $m = 0, 1, 2, \dots$. If $b_1 > b_3 + b_4\tau$ and $\xi = \eta - (b_1 + b_2)\theta > 0$, then $y(t) \leq \sup_{-\tau \leq v \leq 0} \tilde{h}(v) \exp(-\xi t)$, where $\eta > 0$ is the solution of $\eta - b_1 + b_3 \exp\{\eta\tau\} - \frac{b_4}{\eta}(1 - \exp\{\eta\tau\}) = 0$.

3 Main results

We firstly pay attention to the network synchronization under aperiodically intermittent control by employing the matrix measure approach and then extend the derived results to switched dynamical networks with

unstable modes by designing a feasible switching rule with proper MDADT. In the first place, we derive some novel synchronization criteria for (1) by utilizing the ω -measure approach and designing an efficient aperiodically intermittent feedback scheme.

Theorem 1 With Assumptions 1 and 2, (1) will achieve GES under (4), if there are $\xi > 0$ and $\eta > 0$ satisfying

- (1) $a_1 > \|Z_2\|_\omega + a_2 + a_3\tau > 0$;
- (2) $\mu_\omega(c_1(D \otimes \Gamma_1) + Z_1 \otimes I_n) > 0$;
- (3) $\xi = \eta - (a_1 + a'_1)\theta > 0$,

where $a_1 = -\mu_\omega(c_1(D \otimes \Gamma_1) + (Z_1 - U) \otimes I_n)$, $a_2 = c_2 \|A \otimes \Gamma_2\|_\omega$, $a_3 = c_3 \|B \otimes \Gamma_3\|_\omega$, $a'_1 = \mu_\omega(c_1(D \otimes \Gamma_1) + Z_1 \otimes I_n)$, $Z_1 = \text{diag}(z_{11}, z_{12}, \dots, z_{1N})$, $Z_2 = \text{diag}(z_{21}, z_{22}, \dots, z_{2N})$, $\eta > 0$ is the unique positive solution of $\eta - a_1 + (\|Z_2\|_\omega + a_2) \exp\{\eta\tau\} - \frac{a_3}{\eta}(1 - \exp\{\eta\tau\}) = 0$.

Proof Differentiating $\|\tilde{h}(t)\|_\omega$ along (5) and taking the first-order Taylor expansion, we derive the following results. When $t_m \leq t < h_m$, $m = 0, 1, 2, \dots$, one has

$$\begin{aligned} & \lim_{\chi \rightarrow 0^+} \frac{\|\tilde{h}(t + \chi)\|_\omega - \|\tilde{h}(t)\|_\omega}{\chi} \\ &= \lim_{\chi \rightarrow 0^+} \frac{1}{\chi} \left\{ \|\tilde{h}(t) + \chi[\tilde{H}(\tilde{h}(t)) + c_1(D \otimes \Gamma_1)\tilde{h}(t) \right. \\ & \quad + c_2(A \otimes \Gamma_2)\tilde{h}(t - \tau_2(t)) \\ & \quad + c_3(B \otimes \Gamma_3) \int_{t-\tau_3(t)}^t \tilde{h}(v) dv - U\tilde{h}(t)] \\ & \quad \left. + o(\chi)\|_\omega - \|\tilde{h}(t)\|_\omega \right\}. \end{aligned}$$

Considering $\tilde{H}(\tilde{h}(t)) \leq Z_1\tilde{h}(t) + Z_2\tilde{h}(t - \tau_1(t))$ from Assumption 1, we know

$$\begin{aligned} D^+ \|\tilde{h}(t)\|_\omega & \leq \lim_{\chi \rightarrow 0^+} \left\{ \frac{\|I + \chi(c_1(D \otimes \Gamma_1) + (Z_1 - U) \otimes I_n)\|_\omega - 1}{\chi} \|\tilde{h}(t)\|_\omega \right. \\ & \quad + \|Z_2\tilde{h}(t - \tau_1(t)) + c_2(A \otimes \Gamma_2)\tilde{h}(t - \tau_2(t))\|_\omega \\ & \quad \left. + \|c_3(B \otimes \Gamma_3) \int_{t-\tau_3(t)}^t \tilde{h}(v) dv\|_\omega \right\} \\ & \leq \mu_\omega(c_1(D \otimes \Gamma_1) + (Z_1 - U) \otimes I_n) \|\tilde{h}(t)\|_\omega \\ & \quad + (\|Z_2\|_\omega + c_2 \|A \otimes \Gamma_2\|_\omega) \sup_{t-\tau \leq v \leq t} \|\tilde{h}(v)\|_\omega \\ & \quad + c_3 \|B \otimes \Gamma_3\|_\omega \int_{t-\tau_3(t)}^t \|\tilde{h}(v)\|_\omega dv. \end{aligned}$$

Let $a_1 = -\mu_\omega(c_1(D \otimes \Gamma_1) + (Z_1 - U) \otimes I_n)$, $a_2 = c_2 \|A \otimes \Gamma_2\|_\omega$ and $a_3 = c_3 \|B \otimes \Gamma_3\|_\omega$, we derive

$$D^+ \|\tilde{h}(t)\|_\omega \leq -a_1 \|\tilde{h}(t)\|_\omega$$

$$\begin{aligned}
& + (\|Z_2\|_\omega + a_2) \sup_{t-\tau \leq v \leq t} \|\tilde{h}(v)\|_\omega \\
& + a_3 \int_{t-\tau_3(t)}^t \|\tilde{h}(v)\|_\omega dv. \quad (9)
\end{aligned}$$

When $h_m \leq t < t_{m+1}$, $m = 0, 1, 2, \dots$, we repeat similar derivative calculation and set $a'_1 = \mu_\omega(c_1(D \otimes \Gamma_1) + Z_1 \otimes I_n)$, and we get

$$\begin{aligned}
D^+ \|\tilde{h}(t)\|_\omega & \leq a'_1 \|\tilde{h}(t)\|_\omega \\
& + (\|Z_2\|_\omega + a_2) \sup_{t-\tau \leq v \leq t} \|\tilde{h}(v)\|_\omega \\
& + a_3 \int_{t-\tau_3(t)}^t \|\tilde{h}(v)\|_\omega dv. \quad (10)
\end{aligned}$$

Combining (9) and (10), we derive from Lemma 1 and conditions in Theorem 1 that

$$\begin{aligned}
\|\tilde{h}(t)\|_\omega & \leq \sup_{-\tau \leq v \leq 0} \|\tilde{h}(v)\|_\omega \exp\{-\xi t\}, \quad \xi > 0, \quad t \\
& \geq -\tau. \quad (11)
\end{aligned}$$

Thus, we directly draw a conclusion that (1) globally exponentially synchronize to $\varphi(t)$ under (4). $\square$

If the internal time delay is not involved in complex network (1), i.e., $\tau_1(t) = 0$, which also implies $z_{2i} = 0$ in Assumption 1, then by similar inference process, the following Corollaries are obtained consequently.

Corollary 1 *With Assumptions 1 and 2, (1) will achieve GES if (2) and (3) in Theorem 1 and the following condition hold:*

$$a_1 > a_2 + a_3\tau > 0, \quad (12)$$

where $\eta > 0$ and $\eta - a_1 + a_2 \exp\{\eta\tau\} - \frac{a_3}{\eta}(1 - \exp\{\eta\tau\}) = 0$.

Similarly, for complex network (1) without distributed coupling time delay, i.e., $\tau_3(t) = 0$, Corollary 2 can be derived as follows:

Corollary 2 *With Assumptions 1 and 2, (1) will realize GES if (2) and (3) in Theorem 1 and*

$$a_1 > \|Z_2\|_\omega + a_2 > 0 \quad (13)$$

hold, where $\eta > 0$ and $\eta - a_1 + (\|Z_2\|_\omega + a_2) \exp\{\eta\tau\} = 0$.

When $h_m - t_m \equiv \delta$, $t_{m+1} - t_m \equiv \varrho$, the aperiodically intermittent control scheme is reduced to periodically intermittent control and the proportion of rest span is $\theta \equiv 1 - \frac{\delta}{\varrho}$. Then, Corollary 3 can be obtained consequently.

Corollary 3 *With Assumptions 1 and 2, (1) will realize GES if conditions (1) and (2) in Theorem 1 and*

$$\xi = \eta - (a_1 + a'_1)(1 - \frac{\delta}{\varrho}) > 0 \quad (14)$$

hold, where $\eta > 0$ and $\eta - a_1 + (\|Z_2\|_\omega + a_2) \exp\{\eta\tau\} - \frac{a_3}{\eta}(1 - \exp\{\eta\tau\}) = 0$.

Remark 2 To our knowledge, this is the first time that we introduce ω -measure to complex dynamical networks and investigate GES under aperiodically intermittent control, which generalizes and enhances the existing results [25–27, 29, 30, 49] to some extent.

Remark 3 According to the definition of p -measure $\mu_p(W) = \lim_{\chi \rightarrow 0^+} \frac{\|I_N + \chi W\|_{p-1}}{\chi}$ and some calculations, we easily obtain $\mu_1(W) = \max_j \{w_{jj} + \sum_{i=1, i \neq j}^N |w_{ij}|\}$ [32]. It is obvious that 1-measure is just a specific situation of ω -measure when $\omega_i = 1$ ($i = 1, 2, \dots, N$).

Next, AIPC and MDADT technique will be employed to investigate GES of SDCNs with unstable modes. For simplicity of representation, we denote

$$V(t) = \|\tilde{h}(t)\|_\omega, \quad t \geq -\tau. \quad (15)$$

Analogously, inequalities (16) are derived from (11).

$$V(t) \leq \sup_{-\tau \leq v \leq 0} V(v) \exp\{-\xi t\}. \quad (16)$$

Theorem 2 *Suppose that $0 \leq \tau \leq \inf_{\kappa} \{t_{\kappa+1} - t_{\kappa}\}$. If for any $p \in \Lambda = \{1, 2, \dots, Q\}$, there are $\xi_p > 0$ and $\eta_p > 0$ satisfying*

- (1) $a_{1p} > \|Z_{2p}\|_\omega + a_{2p} + a_{3p}\tau > 0$;
- (2) $\mu_\omega(c_{1p}(D_p \otimes \Gamma_{1p}) + Z_{1p} \otimes I_n) > 0$;
- (3) let $\lambda_p > 1$, for $\forall (\sigma(t_{\kappa}) = p, \sigma(t_{\kappa}^-) = q) \in \Lambda \times \Lambda$, $p \neq q$,

$$V_p(t_{\kappa}) \leq \lambda_p V_q(t_{\kappa}^-), \quad (17)$$

where $a_{1p} = -\mu_\omega(c_{1p}(D_p \otimes \Gamma_{1p}) + (Z_{1p} - U_p) \otimes I_n)$, $a_{2p} = c_{2p} \|A_p \otimes \Gamma_{2p}\|_\omega$, $a_{3p} = c_{3p} \|B_p \otimes \Gamma_{3p}\|_\omega$, $a'_{1p} = \mu_\omega(c_{1p}(D_p \otimes \Gamma_{1p}) + Z_{1p} \otimes I_n)$, $Z_{1p} = \text{diag}(z_{11p}, \dots, z_{1Np})$ and $Z_{2p} = \text{diag}(z_{21p}, \dots, z_{2Np})$, $\eta_p > 0$ is solution to $\eta_p - a_{1p} + (\|Z_{2p}\|_\omega + a_{2p}) \exp\{\eta_p\tau_p\} - \frac{a_{3p}}{\eta_p}(1 - \exp\{\eta_p\tau_p\}) = 0$, then (6) will achieve globally exponential synchronization as long as $\sigma(t)$ with MDADT satisfying

$$\begin{cases} \tau_{ap} \geq \frac{\ln \lambda_p + \ln \gamma_p}{\xi_p}, & (\xi_p > 0, p \in R), \\ \tau_{ap} \geq \tau_{ap}^* > 0, & (\xi_p < 0, p \in T), \\ \frac{T^-}{T^+} > \frac{\beta^+ + \beta^*}{\beta^- - \beta^*}, & (0 < \beta^* < \beta^-), \end{cases} \quad (18)$$

where $\xi_p = \eta_p - (a_{1p} + a'_{1p})\theta_p$, τ_{ap}^* can be any positive constants, $-\beta^- = \max_{p \in R}(\frac{\ln \lambda_p + \ln \gamma_p}{\tau_{ap}} - \xi_p)$, $\beta^+ = \max_{p \in T}(\frac{\ln \lambda_p + \ln \gamma_p}{\tau_{ap}} - \xi_p)$, and γ_p ($\gamma_p \geq 1$) is a positive constant satisfying $\sup_{t_k - \tau \leq v \leq t_k} V_p(v) \leq \gamma_p V_p(t_k)$. Here, $T^- = \sum_{p=1}^l T_p(t, 0)$ and $T^+ = \sum_{p=l+1}^Q T_p(t, 0)$.

Proof We first activate the p th subsystem within $t \in [t_k, t_{k+1})$. In light of (16), one has

$$\begin{aligned} V_{\sigma(t)}(t) &\leq \sup_{t_k - \tau \leq v \leq t_k} V_{\sigma(t_k)}(v) \exp\{-\xi_{\sigma(t_k)}(t - t_k)\} \\ &\leq \gamma_p V_p(t_k) \exp\{-\xi_p(t - t_k)\}. \end{aligned}$$

Combining equation (17) yields

$$\begin{aligned} V_{\sigma(t)}(t) &\leq \gamma_{\sigma(t_k)} \lambda_{\sigma(t_k)} \exp\{-\xi_{\sigma(t_k)}(t - t_k)\} V_{\sigma(t_k^-)}(t_k^-) \\ &\leq \gamma_{\sigma(t_k)} \gamma_{\sigma(t_{k-1})} \lambda_{\sigma(t_k)} \exp\{-\xi_{\sigma(t_k)}(t - t_k) \\ &\quad - \xi_{\sigma(t_{k-1})}(t_k - t_{k-1})\} V_{\sigma(t_{k-1})}(t_{k-1}) \\ &\leq \gamma_{\sigma(t_k)} \gamma_{\sigma(t_{k-1})} \lambda_{\sigma(t_k)} \lambda_{\sigma(t_{k-1})} \exp\{-\xi_{\sigma(t_k)}(t - t_k) \\ &\quad - \xi_{\sigma(t_{k-1})}(t_k - t_{k-1})\} V_{\sigma(t_{k-1}^-)}(t_{k-1}^-) \\ &\vdots \\ &\leq \gamma_{\sigma(t_k)} \lambda_{\sigma(t_k)} \gamma_{\sigma(t_{k-1})} \lambda_{\sigma(t_{k-1})} \cdots \gamma_{\sigma(t_1)} \lambda_{\sigma(t_1)} \\ &\quad \exp\{-\xi_{\sigma(t_k)}(t - t_k) - \xi_{\sigma(t_{k-1})}(t_k - t_{k-1}) \\ &\quad - \cdots - \xi_{\sigma(t_0)}(t_1 - t_0)\} V_{\sigma(t_0)}(t_0). \end{aligned}$$

Further considering the definition of MDADT, we obtain

$$\begin{aligned} V_{\sigma(t)}(t) &\leq \prod_{p=1}^Q \gamma_p^{(N_{0p} + \frac{T_p(t,0)}{\tau_{ap}})} \prod_{p=1}^l \lambda_p^{(N_{0p} + \frac{T_p(t,0)}{\tau_{ap}})} \\ &\quad \prod_{p=l+1}^Q \lambda_p^{(N_{0p} + \frac{T_p(t,0)}{\tau_{ap}})} \exp\left\{-\sum_{p=1}^l \xi_p T_p(t, 0) \right. \\ &\quad \left. - \sum_{p=l+1}^Q \xi_p T_p(t, 0)\right\} V_{\sigma(0)}(0) \\ &= \exp\left\{\sum_{p=1}^Q N_{0p}(\ln \lambda_p + \ln \gamma_p)\right\} \end{aligned}$$

$$\begin{aligned} &\exp\left\{\sum_{p=1}^l \left(\frac{\ln \lambda_p + \ln \gamma_p}{\tau_{ap}} - \xi_p\right) T_p(t, 0) \right. \\ &\quad \left. + \sum_{p=l+1}^Q \left(\frac{\ln \lambda_p + \ln \gamma_p}{\tau_{ap}} - \xi_p\right) T_p(t, 0)\right\} V_{\sigma(0)}(0). \end{aligned}$$

According to (18), one can verify that

$$\begin{aligned} V_{\sigma(t)}(t) &\leq \exp\left\{\sum_{p=1}^Q N_{0p}(\ln \lambda_p + \ln \gamma_p)\right\} \\ &\quad \exp\left\{-\beta^- T^- + \beta^+ T^+\right\} V_{\sigma(0)}(0) \\ &\leq \exp\left\{\sum_{p=1}^Q N_{0p}(\ln \lambda_p + \ln \gamma_p)\right\} \\ &\quad \exp\left\{-\beta^*(T^- + T^+)\right\} V_{\sigma(0)}(0). \end{aligned}$$

Set $M = \exp\left\{\sum_{p=1}^Q N_{0p}(\ln \lambda_p + \ln \gamma_p)\right\}$. Consequently, we have

$$V_{\sigma(t)}(t) \leq M \exp\left\{-\beta^* t\right\} V_{\sigma(0)}(0).$$

By (15), we draw a conclusion that (6) with unstable modes achieves GES via AIPC provided MDADT of (18) holds. $\square$

Remark 4 Notably, AIPC is firstly utilized to synchronize SDCNs by designing a feasible switching rule with proper MDADT. Furthermore, a more meticulous work is conducted by taking time-varying delays into account when the system switches between different subsystems, which also shows the contribution and improvement over the previous results [28, 49].

4 Simulation examples and analysis

Consider SDCNs (6) consisting of two subsystems, where $x_i(t)$ is a two-dimensional column vector, $\sigma(t) : [0, \infty) \rightarrow \Lambda = \{1, 2\}$, $f_1(\cdot) = f_2(\cdot)$, and each subsystem consists of 12 identical nodes. In general, we suppose subsystem 1 is stable and subsystem 2 is unstable. For subsystem 1, $c_{11} = 10$, $c_{21} = 1$, $c_{31} = 4$, the inner connecting matrices are

$$\Gamma_{11} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \Gamma_{21} = \begin{bmatrix} 0.1 & 0.3 \\ -0.1 & 0.2 \end{bmatrix}, \Gamma_{31} = \begin{bmatrix} 0.2 & 0.1 \\ 0.3 & 0.5 \end{bmatrix},$$

and the coupling configuration matrices are obtained as

$$\begin{aligned}
D1 &= \begin{pmatrix} -6 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & -6 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & -7 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -6 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & -6 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & -5 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & -4 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 & -5 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & -4 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & -6 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & -5 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & -4 \end{pmatrix}, \\
A1 &= \begin{pmatrix} -6 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & -7 & 1 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & -7 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -6 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & -6 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & -5 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & -3 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & -8 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & -4 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & -6 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & -5 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & -4 \end{pmatrix}, \\
B1 &= \begin{pmatrix} -5 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & -4 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & -4 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -2 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & -5 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & -4 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & -5 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & -4 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & -3 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & -4 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -3 \end{pmatrix}.
\end{aligned}$$

For unstable subsystem 2, we select $c_{12} = 25$, $c_{22} = 1$, $c_{32} = 5$ and

$$\Gamma_{12} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \Gamma_{22} = \begin{bmatrix} 0.1 & 0.2 \\ -0.1 & 0.1 \end{bmatrix}, \Gamma_{32} = \begin{bmatrix} 0.3 & 0.2 \\ 0.1 & 0.5 \end{bmatrix}.$$

The coupling configuration matrices are, respectively,

$$D2 = \begin{pmatrix} -8 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & -7 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & -6 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & -7 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & -9 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 & -6 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & -5 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 1 & 0 & -6 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & -8 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & -5 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & -7 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & -4 \end{pmatrix},$$

$$\begin{aligned}
A2 &= \begin{pmatrix} -4 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & -3 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -5 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -3 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & -3 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & -2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & -4 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & -2 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & -4 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & -2 \end{pmatrix}, \\
B2 &= \begin{pmatrix} -3 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -3 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & -2 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & -3 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -3 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -2 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & -6 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & -3 \end{pmatrix}.
\end{aligned}$$

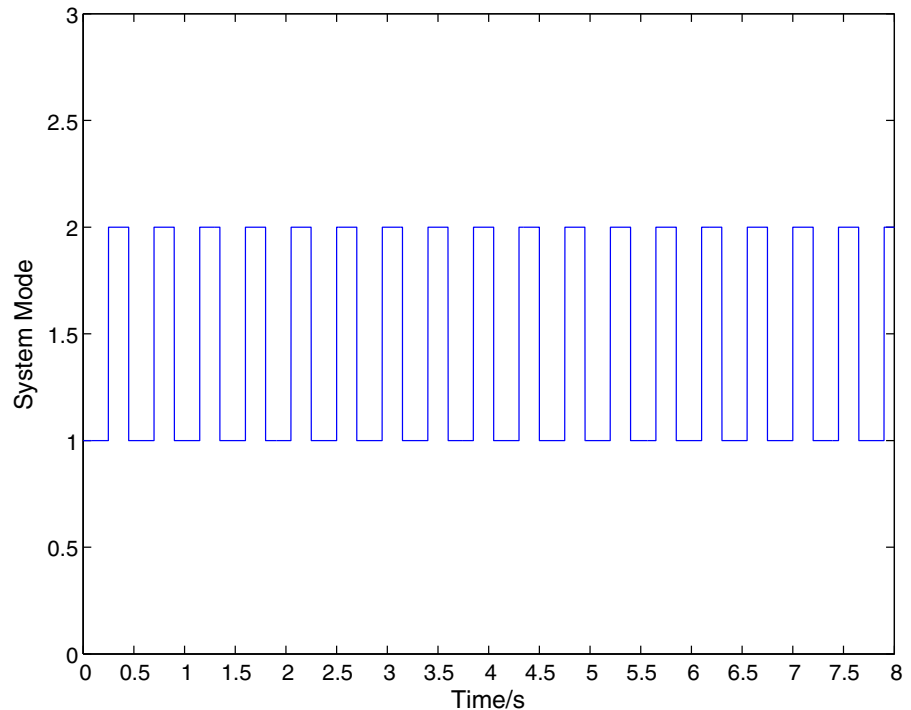
Example 1 Consider function $f(\cdot)$ in (6) as

$$\begin{aligned}
\tilde{f}(x_i(t)) &= -C_1 x_i(t) + C_2 g(x_i(t)) \\
&\quad + C_3 g(x_i(t - \tau_1(t))),
\end{aligned} \tag{19}$$

where $C_1 = I_2$, $C_2 = \begin{bmatrix} 2.0 & -0.5 \\ -3.0 & 5.0 \end{bmatrix}$, $C_3 = \begin{bmatrix} -2.0 & -0.1 \\ 0.5 & 4.0 \end{bmatrix}$, and we choose $g(x_i(t)) = (\tanh(x_{i1}(t)), \tanh(x_{i2}(t)))^T$. By calculation, we take $z_{1i} = 16$ and $z_{2i} = 0.5$ ($i = 1, 2, \dots, 12$) such that Assumption 1 holds. For simplicity, we set $\tau_1(t) = 0.1s$, $\tau_2(t) = 0.1s$, $\tau_3(t) = 0.08s$ and $\tau = \max\{\tau_1, \tau_2, \tau_3\} = 0.1s$. Moreover, for $v \in [-\tau, 0]$, $\varphi(v) = (0.5, 0.6)^T$, $x_{i1}(v) = 0.5 + 0.5i$ and $x_{i2}(v) = 0.6 + 0.5i$ ($i = 1, 2, \dots, 5$), $x_{i1}(v) = 0.5 - 0.5i$ and $x_{i2}(v) = 0.6 - 0.5i$ ($i = 6, 7, \dots, 12$).

For subsystem 1, $U_1 = \text{diag}(20, 10, 15, 8, 12, 0, \dots, 0)$ and the number of pinning nodes is 5. Then, we obtain $a_{11} = 15.5714$, $a_{21} = 1.585$, $a_{31} = 1.056$, $a'_{11} = 2$. Solving $\eta_1 - a_{11} + (\|Z_{21}\|_\omega + a_{21}) \exp\{\eta_1 \tau_1\} - \frac{a_{31}}{\eta_1} (1 - \exp\{\eta_1 \tau_1\}) = 0$ yields $\eta_1 = 10.6531$. Take $\theta_1 = 0.5$, and it follows that $\xi_1 = \eta_1 - (a_{11} + a'_{11})\theta_1 = 1.8674 > 0$. Let $\lambda_1 = 1.4$, $\gamma_1 = 1.2$, then $\tau_{a1} \geq \frac{\ln \lambda_1 + \ln \gamma_1}{\xi_1} = 0.2778$, and we choose the average dwell time of subsystem 1 as $\tau_{a1} = 0.5s$. Thus, $-\beta^- = \frac{\ln \lambda_1 + \ln \gamma_1}{\tau_{a1}} - \xi_1 = -0.8298$,

Fig. 1 Switching modes of signal for Example 1



$\beta^- = 0.8298$, and we take $\beta^- > \beta^* = 0.01$. The aperiodically intermittent controller works on time interval $[0, 0.1) \cup [0.18, 0.3) \cup [0.4, 0.43)$.

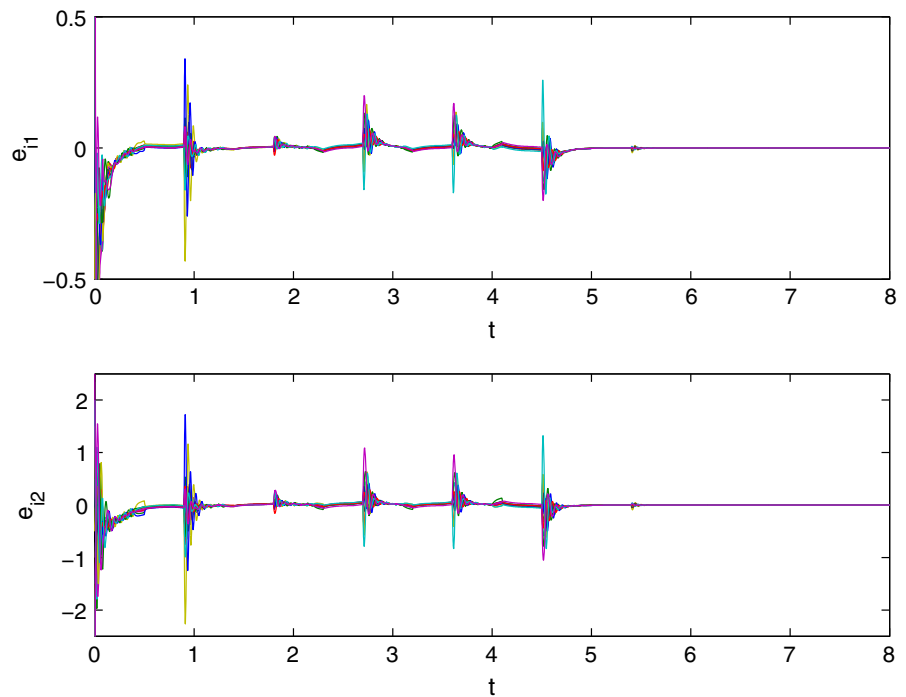
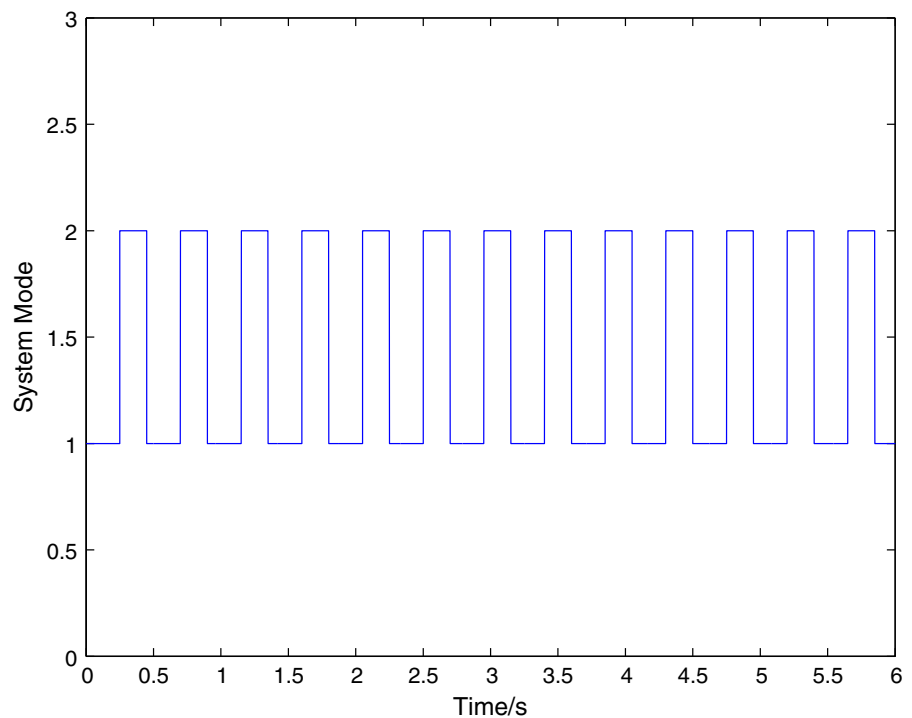
For subsystem 2, the control gain matrix is $U_2 = \text{diag}(40, 15, 25, 10, 20, 0, \dots, 0)$. By calculation, we derive that $a_{12} = 29$, $a_{22} = 1.23$, $a_{32} = 0.4$, $a'_{12} = 11$, $\eta_2 = 19.698$. Set $\theta_2 = 0.5$, then $\xi_2 = \eta_2 - (a_{12} + a'_{12})\theta_2 = -0.302 < 0$. Choosing $\lambda_2 = 1.2$, $\gamma_2 = 1.1$ and $\tau_{a2} = 0.4s$, it follows that $\beta^+ = \frac{\ln \lambda_2 + \ln \gamma_2}{\tau_{a2}} - \xi_2 = 0.9961$. The intermittent controller works on time span $[0.5, 0.54) \cup [0.6, 0.71) \cup [0.8, 0.86)$. As a result, we have $\frac{5}{4} = \frac{T^-}{T^+} > \frac{\beta^+ + \beta^*}{\beta^- - \beta^*} = \frac{1.1061}{0.8198}$. It is quite clear that the switched delayed complex network with unstable mode can realize synchronization. Figure 1 shows the system modes under the control of $\sigma(t)$. The dynamical behavior of synchronization error is demonstrated in Fig. 2.

Example 2 Consider the following function with the isolated node,

$$\begin{aligned} \tilde{f}(x_i(t)) = & -H_1 x_i(t) + H_2 g(x_i(t)) \\ & + H_3 g(x_i(t - \tau_1(t))), \end{aligned} \quad (20)$$

where $H_1 = I_2$, $H_2 = \begin{bmatrix} 1 + \frac{\pi}{4} & 20 \\ 0.1 & 1 + \frac{\pi}{4} \end{bmatrix}$, $H_3 = \begin{bmatrix} -1.3\sqrt{2}\pi/4 & 0.1 \\ 0.1 & -1.3\sqrt{2}\pi/4 \end{bmatrix}$, $g(x_i(t)) = \frac{1}{2}(|x_{i1}(t) + 1| - |x_{i1}(t) - 1|, |x_{i2}(t) + 1| - |x_{i2}(t) - 1|)^T$. By calculation, we get $z_{1i} = 21.79$ and $z_{2i} = 1.55$ such that Assumption 1 holds. Time-varying delays are set as $\tau_1(t) = 0.05 \frac{e^t}{1+e^t}$, $\tau_2(t) = 0.05 \frac{e^t}{1+e^t}$, $\tau_3(t) = 0.04 \frac{e^t}{1+e^t}$, and $\tau = \max\{\tau_1, \tau_2, \tau_3\} = 0.05$. Moreover, for $v \in [-\tau, 0]$, $\varphi(v) = (1, 1.2)^T$, $x_{i1}(v) = 1 + 0.5i$ and $x_{i2}(v) = 1.2 + 0.5i$ ($i = 1, 2, \dots, 12$). The number of pinning nodes for the switched dynamical network is 9, and we design the control gain matrices U_i ($i = 1, 2$) as $U_1 = \text{diag}(25, 25, \dots, 25, 0, 0, 0)$, $U_2 = \text{diag}(36, 36, \dots, 36, 0, 0, 0)$.

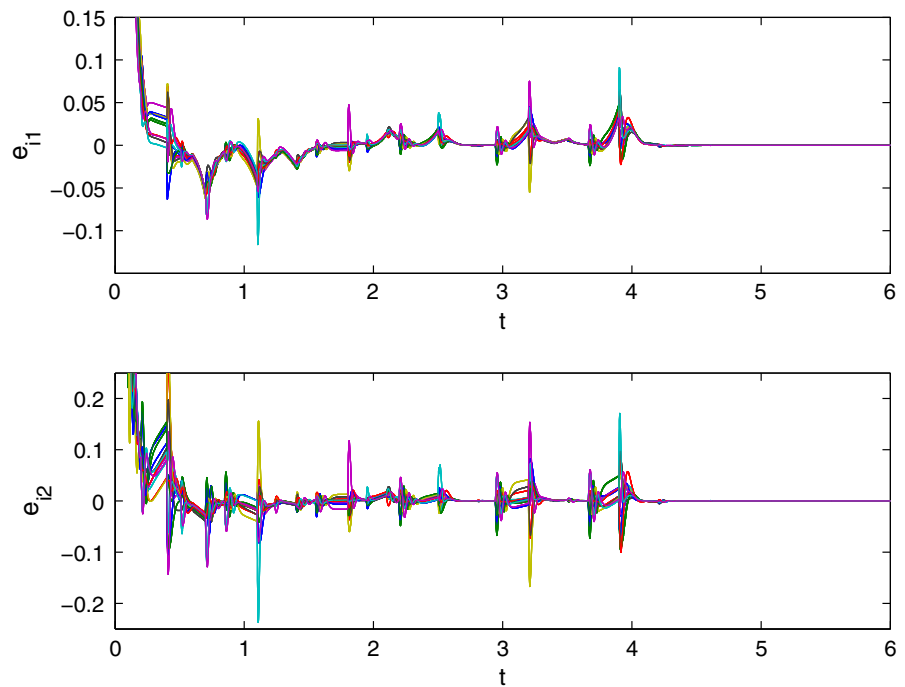
For stable subsystem 1, we obtain by calculation that $a_{11} = 15.71$, $a_{21} = 1.585$, $a_{31} = 1.056$, $a'_{11} = 7.79$, $\eta_1 = 12.402$. Set $\theta_1 = 0.45$, then $\xi_1 = \eta_1 - (a_{11} + a'_{11})\theta_1 = 1.827 > 0$. Let $\lambda_1 = 1.2$, $\gamma_1 = 1.3$, it follows that $\tau_{a1} \geq \frac{\ln \lambda_1 + \ln \gamma_1}{\xi_1} = 0.2434$, and we choose $\tau_{a1} = 0.4s$. Hence, it is derived that $-\beta^- = \frac{\ln \lambda_1 + \ln \gamma_1}{\tau_{a1}} - \xi_1 = -0.9376$, $\beta^- = 0.9376$, and we take $\beta^- > \beta^* = 0.02$. The aperiodically intermittent controller works on time span $[0, 0.06) \cup [0.1, 0.19) \cup [0.25, 0.34)$.

Fig. 2 Time response of $e(t)$ for Example 1**Fig. 3** Switching modes of signal for Example 2

For unstable subsystem 2, we can obtain $a_{12} = 19.21$, $a_{22} = 1.23$, $a_{32} = 0.4$, $a'_{12} = 16.79$, $\eta_2 = 15.8462$. Let $\theta_2 = 0.45$, then $\xi_2 = \eta_2 - (a_{12} + a'_{12})\theta_2 =$

$-0.3538 < 0$. Choose $\lambda_2 = 1.1$, $\gamma_2 = 1.2$ and $\tau_{a2} = 0.3s$, it follows that $\beta^+ = \frac{\ln \lambda_2 + \ln \gamma_2}{\tau_{a2}} - \xi_2 = 1.0479$. The aperiodically intermittent controller is imposed on

Fig. 4 Time response of $e(t)$ for Example 2



time interval $[0.4, 0.455) \cup [0.5, 0.56) \cup [0.6, 0.658)$. Consequently, we derive that $\frac{4}{3} = \frac{T^-}{T^+} > \frac{\beta^+ + \beta^*}{\beta^- - \beta^*} = \frac{1.0479 + 0.02}{0.9376 - 0.02} = \frac{1.0679}{0.9176}$.

It is quite clear that all the criteria in Theorem 2 are satisfied. Therefore, the switched dynamical networks with unstable mode are GES with the desired trajectory. Figure 3 demonstrates the switching modes of the system signal $\sigma(t)$. Figure 4 illustrates the evolution of $e(t)$.

5 Conclusion

This paper has investigated GES issue for SDCNs with unstable modes via AIPC and MDADT approach. Without constructing appropriate multiple Lyapunov functions, ω -measure approach is utilized to derive several synchronization criteria for such network with unstable modes, which is more general and applicative in many investigations. Furthermore, the admissible switching rule with proper MDADT and AIPC scheme is devised. And the AIPC strategy greatly reduces the transmitted information and control cost. Finally, simulation analysis ensures the advantages of proposed innovations. In addition, considering the significant role of stochastic Markov switching systems in operation and engineering fields [51, 52], we will consider

the stability or synchronization for Markov switching systems by utilizing MDADT technique in the future research.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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